Education and Affairs

Laura Aguilar Econometrics Project

For this project, I chose to evaluate the question: Are people with more education less likely to have affairs? This question is relevant given that answering it can help us understand if affairs are motivated by just personal circumstances within a marriage, or if institutional factors, such as education, can impact the decision to have one or abstain. To arrive at an answer, I used data of self reported extramarital affairs from the "A Theory of Extramarital Affairs," Journal of Political Economy, a collection of surveys from married individuals reporting whether or not they have had an affair. The data set is made up of 601 observations, with columns coding for 19 variables like education (in years), religion, occupation, whether or not they have kids, etc. I expect to find that women with more education are less likely to have affairs. The rationale that follows my hypothesis is that people that pursue more education are more socially aware of the ethical and moral implications of having an affair, thus deterring them from doing so.

To test the hypothesis, first, we ran a short linear regression using whether or not the individual reported having an affair (affair) (0/1) against years of education (educ). For an additional year of education, this value of affair increases by approximately 0.003477. We found that this coefficient is not statistically significant because it is less than twice the standard error (0.007365). This result indicates that there is no meaningful relationship between years of education and the chances of having an affair. To address omitted variable bias and model the true effect of education on the probability of having an affair, we decided to control for religion. We did this by adding the variable "relig" to the regression that codes for 5 different levels of religiosity from 5 = very relig to 1 = anti. Through this second regression, we found that the "educ" coefficient became smaller (0.002483), indicating that the increasing effect of education on the probability of an affair is smaller than initially explored in the short regression. However, this value was not statistically significant, given that it was less than twice the standard deviation (0.007315).

Furthermore, the summary of the regression also yielded a coefficient for the variable "relig" of -0.048044, suggesting that an additional degree of religiosity decreases the probability of having an affair by that number. This value was statistically significant because it was more than twice the standard error found (0.015054). These results suggested that religion might have a negative effect on the likelihood of having an affair. Both regressions suggest that years of education have no effect on whether or not an individual has an affair. We do see that religiosity might have an effect on marriage; however, because this is an observational study and not an RCT where this treatment can be randomly assigned, we cannot claim causation. Ultimately, the data does not support the hypothesis that education decreases the likelihood of having an affair. The educ coefficients suggest the opposite but, they are not statistically significant, the inverse relationship is not supported. Furthermore, the long regression is far from perfect. Many variables from the data set were not accounted for. Visual representations were made for both short and long regressions and we also observe no significant patterns. There are also issues with using the variable "affair" because it is a binary variable. Ultimately, the coefficients are meaningless because we are interested in seeing whether or not the individual had an affair (0/1). For future reference, it would have been more appropriate to use the variable "naffairs" which codes for the number of affairs an individual has had. Furthermore, the variable "relig" is a bad control given that it interferes with the accuracy of the reporting of the variable "affair" and "naffairs." In a paper titled "Are there Religious Variations in Marital Infidelity?" by Amy B. Burdette, Christopher G. Ellison, Darren E. Sherkat and Kurt A. Gore, they found that participation in church and actively upholding religious beliefs interferes with accurate reporting of the subjects (Burdette 1553).

While an RCT would address the omitted variable bias of this study, it is neither feasible nor ethical to assign levels of education to people. Nevertheless, to answer the question, we could conduct an IV experiment using an instrument such as quarter of birth. This instrument has been employed before when exploring the effect of education. For instance, researchers Angrist & Krueger employed it in their paper "Does Compulsory School Attendance Affect Schooling and Earnings?" to isolate omitted variables and observe the effect of education on earnings (Angrist & Krueger).

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Table 1

```
affairs_reg1 <- lm(affair ~ educ, data = affairs)
summary(affairs_reg1)
```

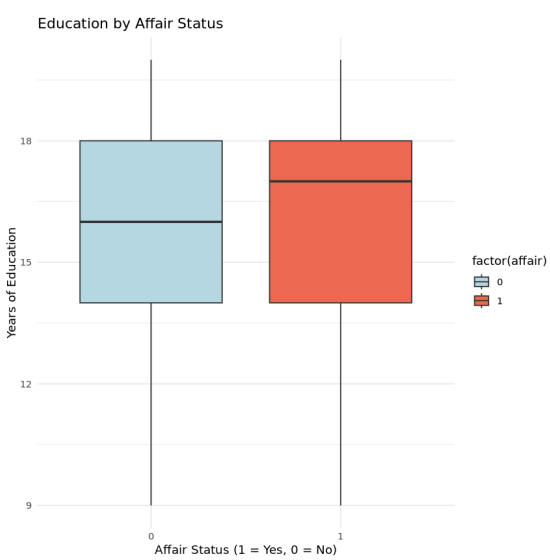
```
Call:
lm(formula = affair ~ educ, data = affairs)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.2629 -0.2525 -0.2420 -0.2247  0.7753 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.193376   0.120366   1.607    0.109
educ        0.003477   0.007365   0.472    0.637

Residual standard error: 0.4334 on 599 degrees of freedom
Multiple R-squared:  0.0003719, Adjusted R-squared:  -0.001297 
F-statistic: 0.2229 on 1 and 599 DF,  p-value: 0.637
```

Table 2

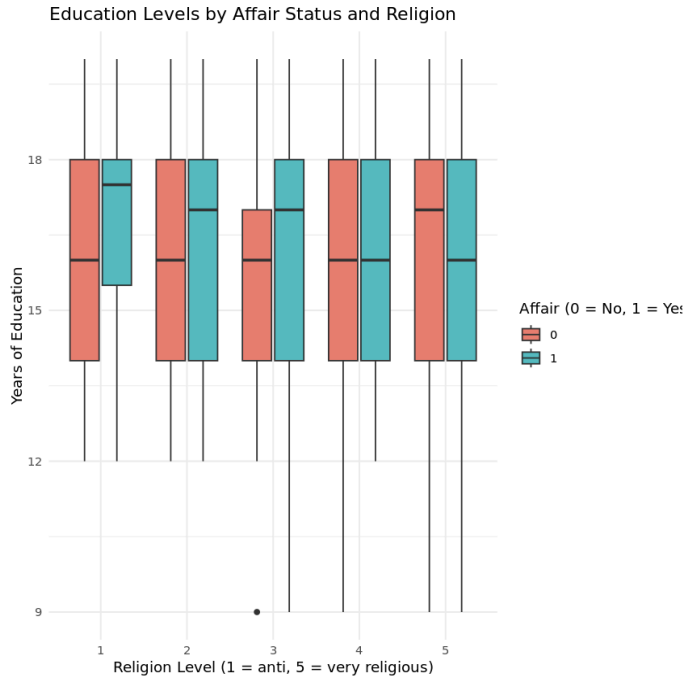
```
affairs_reg2 <- lm(affair ~ educ + relig, data = affairs)
summary(affairs_reg2)
```

```
Call:
lm(formula = affair ~ educ + relig, data = affairs)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.3608 -0.2929 -0.2092 -0.1413  0.8587 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.359173   0.130261   2.757  0.00601 **
educ         0.002483   0.007315   0.339  0.73442   
relig       -0.048044   0.015054  -3.191  0.00149 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4301 on 598 degrees of freedom
Multiple R-squared:  0.01711,	Adjusted R-squared:  0.01383 
F-statistic: 5.206 on 2 and 598 DF,  p-value: 0.005736
```

Box Plot 1



Box plot 2



Works Cited

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